

Results, Tables, and Supplementary Material

Companion to Manuscript v8 — A matrix-high / DCT-low kidney remodeling direction
recurs across
independent mouse spaceflight transcriptomes and aligns with the Cosmic Kidney Disease
NCC phenotype

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Draft v8, May 2026

This document collects the full results tables, all figures, and supplementary material supporting Manuscript v8. It is organized to follow the manuscript's claim hierarchy: cohort design; the original within-RRRM-2 age-rewiring test; cross-cohort flight-vector recurrence; the matrix-high / DCT-low state; the live-return arm; mechanism context (OSD-253); and the pre-registered OSD-462 multi-omic anchor. All numbers were regenerated from the run artifacts `run_20260519_000547_2500g` (transcriptomic) and `run_20260521_osd462_anchor` (multi-omic anchor), and independently spot-verified against the raw OSD-462 workbooks.

Contents

S1 Pre-registration and hypothesis outcomes

The OSD-462 multi-omic anchor was specified in a pre-registered plan (docs/osd462_multiomics_analysis_plan.md) with four hypotheses, each carrying an explicit falsification rule declared before any protein or phosphoprotein effect was computed. Table ?? records the outcomes. Three hypotheses were falsified and one was strongly supported; under the pre-registration, every outcome — including the nulls — is a reportable result.

Table S1: Pre-registered hypotheses and outcomes for the OSD-462 multi-omic anchor.

ID	Hypothesis and pre-registered prediction	Outcome
H1	Matrix/ECM transcripts and proteins are concordant in flight (matrix proteins up; concordance exceeds matched null).	Falsified. Matrix proteins moved opposite transcripts.
H2	DCT/NCC-WNK transcripts and proteins are concordant (transporter proteins down; concordance exceeds null).	Falsified. No protein-abundance effect; total NCC protein flat.
H3	The WNK-SPAK/OSR1-NCC phospho-axis is suppressed in flight.	Supported. NCC activating cluster and SPAK strongly down.
H4	RRRM-2 LIONESS/node2vec network candidates are enriched for OSD-462 protein- or phosphoprotein-changing genes.	Falsified. No enrichment ($p \geq 0.23$).

Decision-table outcome. Protein-abundance concordance was null for both matrix and DCT gene sets; the DCT signal relocated to the phospho/activity layer. Per the pre-registered claim ceiling, the result is a molecular claim (transcript- and signaling-level), with histology and urine/serum chemistry remaining out of scope and explicitly future work.

S2 Cohort design

Table S2: **Cohorts analyzed.** RRRM-2 is the factorial bridge; OSD-513 and OSD-253 are independent transcriptomic cohorts; OSD-462 (RR-10) is the multi-omic anchor and the source of the Cosmic Kidney Disease mouse kidney phosphoproteomics.

Cohort	Accession	Tissue / sex	Modalities	Role
RRRM-2	OSD-771	Kidney / female	RNA-seq	Factorial bridge: Age × Arm (ISS-T, LAR) × Flight, $n = 5/\text{cell}$
OSD-513	OSD-513	Kidney / –	RNA-seq	Independent flight-vector re-currence co-hort
OSD-253	OSD-253	Kidney / –	RNA-seq	Strain/mechanism context (C57BL/6J vs C3H/HeJ)
OSD-462	OSD-462 (RR-10)	Kidney / female	RNA-seq, proteomics, phosphoproteomics	Multi-omic anchor; Cosmic Kidney Disease mouse co-hort
Tabula Muris Senis	—	Kidney	scRNA-seq	External aging reference

S3 Within-RRRM-2 age-vector stability (original claim, not supported)

Table S3: **Bootstrap angular stability of RRRM-2 control-aging vectors.** Module and pathway are the primary biological resolutions; three of four arm-resolution settings failed the stability gate, so the original within-cohort age-rewiring contrast is not promoted as a claim.

Arm	Resolution	Full norm	Median cosine	Lower CI	Required lower	Passed
ISS-T	pathway	0.167	0.787	0.248	0.400	no
ISS-T	module	0.325	0.750	0.056	0.400	no
ISS-T	gene	10.152	0.746	0.545	0.200	yes
ISS-T	LIONESS-node	10.152	0.745	0.549	0.200	yes
LAR	pathway	0.221	0.890	0.345	0.400	no
LAR	module	0.353	0.774	0.493	0.400	yes
LAR	gene	9.687	0.742	0.563	0.200	yes
LAR	LIONESS-node	9.687	0.737	0.556	0.200	yes

S4 Cross-OSDR flight-vector recurrence

Table S4: **Cross-OSDR pathway-vector alignment.** OSD-513 is interpreted as directional recurrence; OSD-253 as context. p_{perm} is the directional external-label permutation sensitivity with the RRRM-2 reference vector held fixed.

Study	Scenario	Arm	Point	Median	Lower CI	Upper CI	p_{perm}
OSD-513	powered HGC	ISS-T	0.641	0.625	0.351	0.814	0.127
OSD-513	powered HGC	LAR	-0.511	-0.498	-0.726	-0.231	0.178
OSD-253	original GC	ISS-T	0.674	0.602	-0.411	0.836	0.141
OSD-253	original GC	LAR	-0.679	-0.608	-0.816	0.415	0.125
OSD-253	rerun GC	ISS-T	0.474	0.470	-0.040	0.688	0.269
OSD-253	rerun GC	LAR	-0.318	-0.309	-0.572	0.183	0.344

Table S5: **Pathway construction of the recurrent ISS-T remodeling axis.** Effects are pathway-scale FLT-minus-GC contrasts; the axis is oriented so ECM/fibrosis is positive and DCT/NCC-WNK suppression is negative.

Pathway	RRRM-2 ISS-T	OSD-513	Axis weight	Primary axis
ECM remodeling	0.949	0.801	0.622	yes
Fibrosis	0.764	0.652	0.504	yes
DCT/NCC-WNK	-0.398	-0.889	-0.452	yes
Apoptosis	0.674	0.386	0.378	yes
Oxidative stress	0.198	0.104	0.108	yes
Inflammation	0.561	-0.092	0.000	no
Ion transport	0.329	-0.762	0.000	sensitivity
Lipid metabolism	-0.320	0.156	0.000	no
Calcium handling	0.003	-0.550	0.000	no

S5 Matrix-high / DCT-low state and live-return arm

Table S6: **State-space component effects (FLT-minus-GC) and arm-by-flight interactions.** Young LAR effects are shown to be underpowered and not statistically distinguishable from zero.

Component	Arm / scope	Effect	Welch p / interaction p	BH q
matrix-minus-DCT	ISS-T pooled	+2.254	0.003	0.020
matrix-minus-DCT	LAR pooled	-0.112	0.865	0.895
matrix-minus-DCT	ISS-T young	+3.204	0.005	0.025
matrix-minus-DCT	LAR young	-0.782	0.361	0.542
matrix-minus-DCT	Arm $\times$ Flight interaction	+2.366	0.016	0.027
matrix component	Arm $\times$ Flight interaction	+1.738	—	0.004
DCT transport	Arm $\times$ Flight interaction	-0.628	—	0.344
mechanism-score vector	cos(LAR, ISS-T) pooled	-0.725	CI [-0.874, -0.122]	—

S6 OSD-462 multi-omic anchor

S6.1 RNA recurrence gate (Layer 4)

The OSD-462 Space Flight minus Ground Control RNA pathway vector reproduced the RRRM-2 ISS-T matrix-high / DCT-low direction with cosine 0.869 (95% bootstrap CI 0.648–0.903; bootstrap median 0.823) across 11 pathways; leave-one-pathway-out cosine ranged 0.823–0.904. The recurrence gate passed.

Table S7: **OSD-462 RNA pathway flight effects vs RRRM-2 ISS-T.** Space Flight minus Ground Control (OSD-462) and FLT-minus-GC (RRRM-2 ISS-T).

Pathway	OSD-462 RNA effect	RRRM-2 ISS-T effect	Sign agreement
ecm_organization	0.222	0.280	1.000
fibrosis_tgfb_empt	0.149	0.025	1.000
tlr4_innate	0.075	0.095	1.000
integrin_cell_adhesion	0.119	0.087	1.000
slp_slpr3	0.123	0.106	1.000
mmp_adam_proteolysis	0.197	0.123	1.000
dct_ncc_wnk_transport	-0.130	-0.075	1.000
tubular_transport_broad	-0.048	0.042	0.000
oxidative_stress_nrf2	0.021	0.005	1.000
macrophage_inflammation	0.109	0.090	1.000
preservation_stress_response	0.157	0.293	1.000

S6.2 Protein-abundance concordance (Layer 1)

Genome-wide RRRM-2-RNA versus OSD-462-protein flight-effect correlation was negligible (Spearman 0.043). No targeted gene set was concordant in its predicted direction against the abundance- and peptide-matched random-gene-set null. The matrix/ECM set moved opposite its transcripts.

Table S8: **OSD-462 protein-abundance concordance by gene set.** Signed mean protein flight effect with the matched random-gene-set null interval. No set is concordant in its predicted direction.

Gene set	Kind	n	Predicted dir.	Signed mean effect	Matched-null interval
dct_ncc_wnk_transport	family	10	down	-0.044	[-0.052, 0.047]
ecm_organization	family	15	up	-0.107	[-0.082, 0.056]
tlr4_innate	family	8	up	-0.021	[-0.040, 0.031]
slp_slpr3	family	9	up	-0.045	[-0.055, 0.054]
tubular_transport_broad	control	15	up	0.043	[-0.061, 0.060]

S6.3 WNK–SPAK/OSR1–NCC phospho-axis (Layer 2)

The phosphoproteomics checkpoint passed (all targeted NCC, SPAK, OSR1, WNK sites quantified). Total NCC protein abundance was unchanged in flight (+0.089), so phosphosite changes reflect altered phosphorylation stoichiometry. Table ?? lists every quantified phosphosite on the WNK–SPAK/OSR1–NCC chain. The NCC N-terminal regulatory (SPAK/OSR1-target) cluster and SPAK sites are strongly suppressed; non-regulatory NCC sites are unchanged — an internal negative control against a global normalization artifact.

Table S9: **Complete OSD-462 WNK–SPAK/OSR1–NCC phosphosite flight effects.** Space Flight minus Ground Control, log₂ scale, plex-corrected. *n* is FL/GC channels.

	Gene	Site	Kind	<i>n</i>	Effect	CI low	CI high	<i>p</i> value
Calb1	60		single		5/5	0.177	-0.016	0.370
Cul3	737		single		10/10	0.121	-4.8e-04	0.243
Nedd4l	331		single		5/5	0.077	-0.223	0.378
Nedd4l	332		single		10/10	0.002	-0.069	0.073
Nedd4l	373		single		10/10	-0.230	-0.440	-0.021
Nedd4l	480		single		10/10	-0.264	-0.383	-0.144
Nedd4l	484		single		5/5	-0.412	-0.637	-0.186
Nedd4l	504		single		10/10	0.040	-0.106	0.186
Oxsr1	339		single		10/10	0.005	-0.055	0.065
Sgk1	494		single		10/10	-0.114	-0.683	0.454
Slc12a3	120		single		10/10	0.027	-0.175	0.229
Slc12a3	120;122		composite		10/10	0.191	-0.046	0.429
Slc12a3	120;124		composite		10/10	0.039	-0.164	0.243
Slc12a3	122		single		10/10	0.045	-0.168	0.257
Slc12a3	122;124		composite		10/10	0.168	-0.062	0.399
Slc12a3	124		single		10/10	0.051	-0.254	0.357
Slc12a3	53		single		10/10	-0.851	-1.257	-0.445
Slc12a3	53;65		composite		10/10	-0.350	-0.496	-0.204
Slc12a3	53;68		composite		5/5	-0.133	-0.758	0.493
Slc12a3	56;65		composite		5/5	-0.124	-0.560	0.312
Slc12a3	58;65		composite		10/10	-0.545	-0.836	-0.254
Slc12a3	65		single		10/10	-0.790	-1.038	-0.543
Slc12a3	65;68		composite		10/10	-1.563	-2.005	-1.121
Slc12a3	68		single		10/10	-0.794	-1.168	-0.420
Slc12a3	89		single		10/10	-0.930	-1.305	-0.556
Slc12a3	96		single		10/10	0.048	-0.092	0.189
Stk39	366		single		10/10	-0.520	-0.706	-0.334
Stk39	382		single		5/5	-0.829	-1.768	0.109
Stk39	382;383		composite		10/10	-0.368	-0.579	-0.157
Stk39	383		single		10/10	-0.793	-1.160	-0.426
Stk39	397		single		10/10	0.073	-0.013	0.159
Wnk1	157		single		5/5	-0.129	-0.346	0.087
Wnk1	17		single		10/10	-0.159	-0.335	0.018
Wnk1	172		single		10/10	0.014	-0.128	0.156
Wnk1	185		single		5/5	0.008	-0.252	0.269
Wnk1	2258		single		10/10	-0.200	-0.482	0.083
Wnk1	2280;2282		composite		5/5	-0.121	-0.592	0.350
Wnk1	2280;2282;2285		composite		10/10	0.129	-0.017	0.275
Wnk1	2282;2285		composite		10/10	0.135	-0.120	0.390
Wnk1	2285		single		5/5	0.022	-0.467	0.510
Wnk1	2392		single		5/5	-0.628	-0.880	-0.376
Wnk1	2623		single		5/5	-0.297	-0.563	-0.030
Wnk1	2625		single		10/10	-0.362	-0.609	-0.114
Wnk4	1014		single		10/10	-0.036	-0.096	0.024
Wnk4	1025		single		10/10	0.190	0.051	0.329
Wnk4	1025;1033		composite		10/10	0.287	0.093	0.481

Table S9 continued

	Gene	Site	Kind	n	Effect	CI low	CI high	p value
	Wnk4	1033		single	10/10	0.290	0.097	0.484
	Wnk4	125		single	10/10	-0.049	-0.273	0.175
	Wnk4	639		single	5/5	-0.110	-0.465	0.246
	Wnk4	783		single	5/5	0.118	-0.033	0.269
	Wnk4	95		single	10/10	-0.006	-0.075	0.063

S6.4 Network-candidate translation (Layer 3)

RRRM-2 LIONESS/node2vec top-composite candidates were not enriched among OSD-462 protein-changing genes at any top-*k* threshold.

Table S10: **Network-candidate translation to OSD-462 proteins.** Enrichment of RRRM-2 network-prioritized genes among protein-changing genes vs the matched null.

Top <i>k</i>	Candidates w/ protein	Mean effect	Null median	Enrichment <i>p</i>
25	14	0.103	0.081	0.227
50	22	0.088	0.079	0.329
100	41	0.089	0.080	0.298

S7 Figures

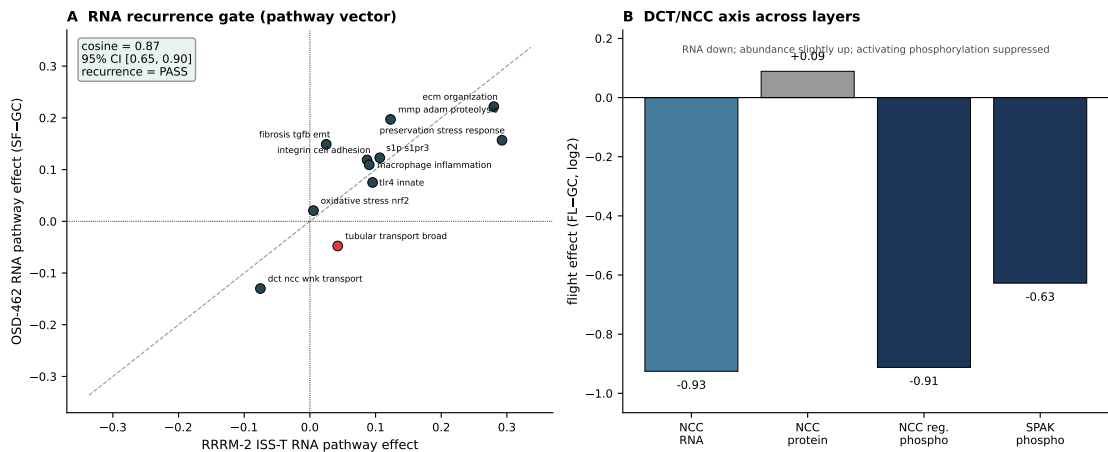


Figure S1: **OSD-462 RNA recurrence gate.** The OSD-462 Space Flight minus Ground Control RNA pathway vector reproduces the RRRM-2 ISS-T matrix-high / DCT-low direction (cosine 0.87, 95% CI 0.65–0.90), with leave-one-pathway-out stability.

OSD-462 / RR-10 multi-omics anchor — protein & phosphoprotein test of the RRRM-2 matrix-high / DCT-low remodeling signal

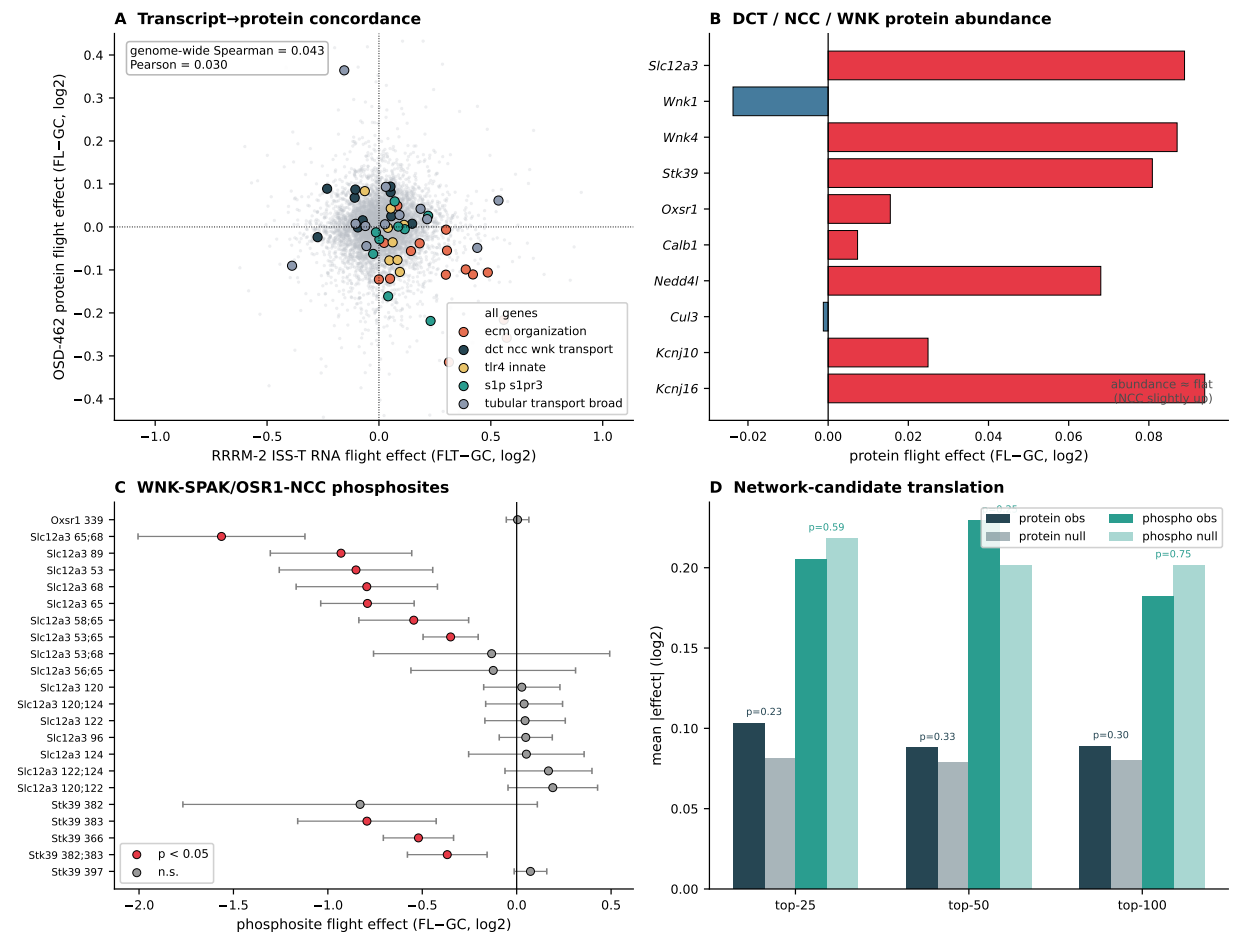


Figure S2: **OSD-462 multi-omic anchor dashboard.** Transcript-level recurrence, protein-abundance null, and phospho-axis suppression of the WNK–SPAK/OSR1–NCC activating cluster.

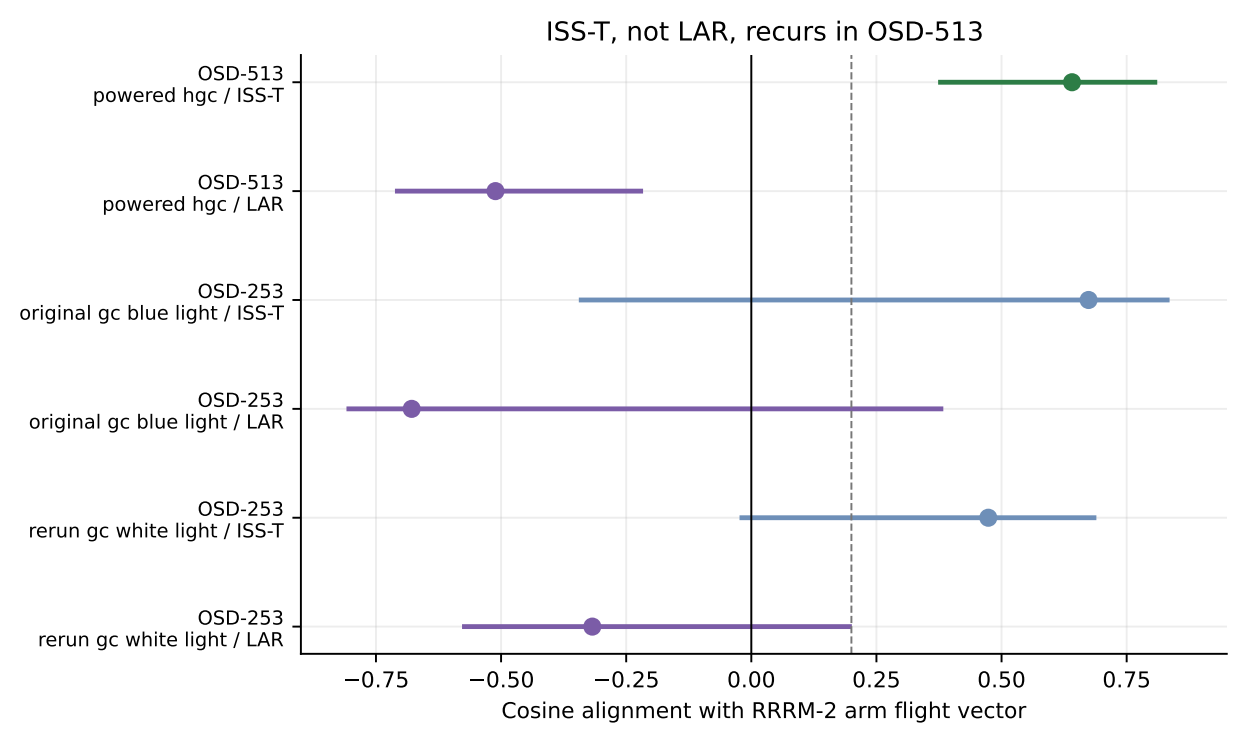


Figure S3: **Cross-OSDR pathway-vector alignment.** RRRM-2 ISS-T aligns with OSD-513; LAR anti-aligns directionally.

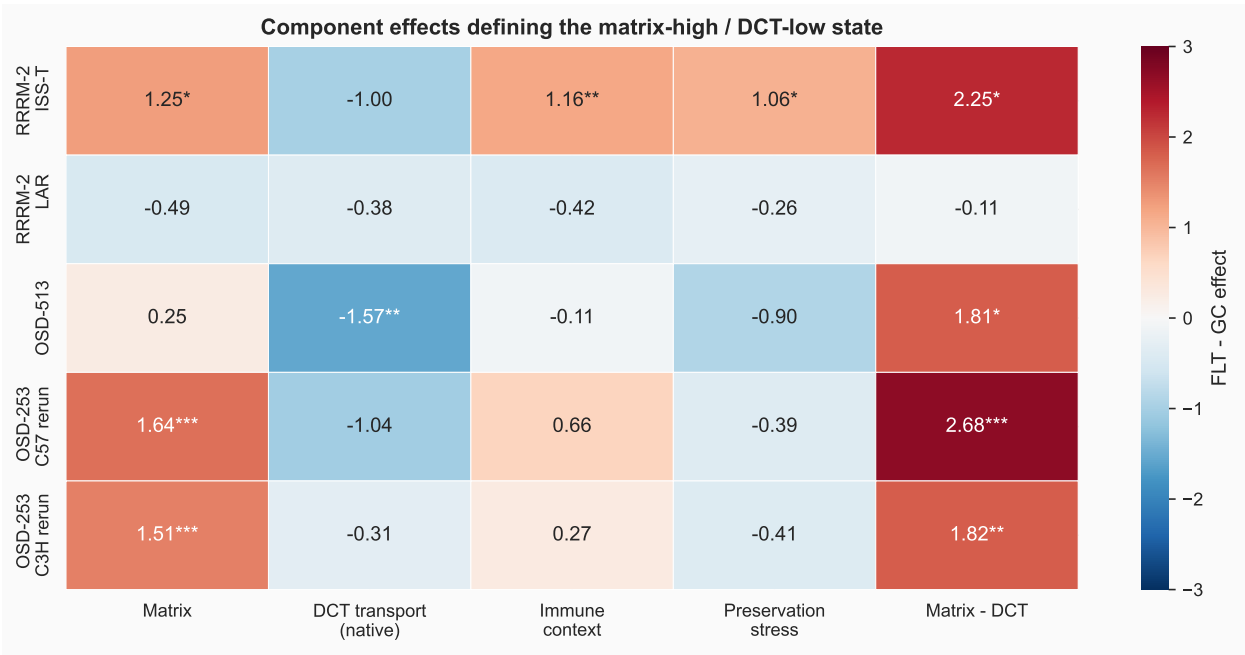


Figure S4: **Matrix-high / DCT-low state component effects.** FLT-minus-GC effects in standardized component units; DCT transport in native direction.

A Gene-level direction check

B Directional geometry by feature layer

	pooled	YNG	OLD
state_space	-0.56	-0.98	-0.38
mechanism_scores	-0.72	-0.77	-0.49
special_clock_s1p_rbm3	-0.27	-0.16	-0.38
wgcna_module_eigengenes	-0.49	-0.39	-0.45
all_gene_expression	-0.19	-0.15	-0.09

C Component effects

D Exploratory mechanism-axis projections

	ISS-T	LAR
circadian_dct_state	-1.94	0.31
immune_tlr4_macrophage	1.24	-0.45
matrix_high_dct_low	1.40	-0.51
preservation_cold_stress	0.40	-0.21
s1p_matrix	1.40	-0.91
transport_rebound	-0.35	-0.91

E Rbm3 / Per2 / S1pr3 by mouse

F Age-stratified state score

Figure S5: **Live-return arm attenuation/divergence dashboard.** The young-LAR opposite point estimate is hypothesis-generating only; the LAR effect is not statistically distinguishable from zero.

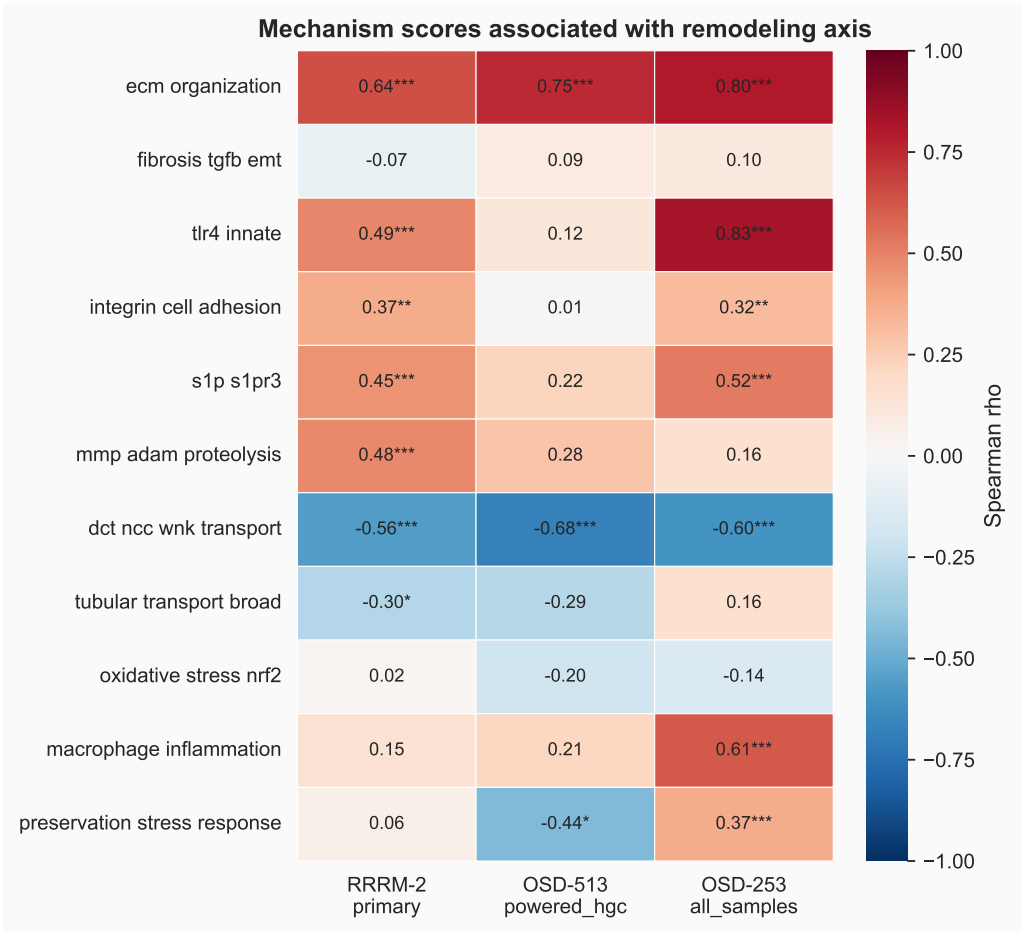


Figure S6: **Mechanism-score associations with the recurrent remodeling score** across RRRM-2, OSD-513, and OSD-253.

S8 Verification notes

The following independent checks were performed on the OSD-462 anchor. (i) The eight `tests/test_osd462_anchor.py` unit tests pass. (ii) NCC phosphosite flight effects were re-derived directly from the raw `siteQuant_360` workbook ($\log_2$ of TMT signal-to-noise, 10 FL vs 10 GC channels, Welch t -test, no pipeline code): Thr53 -0.99 , Thr65 -0.93 ($p = 2.3 \times 10^{-6}$), Thr68 -0.94 , Thr89 -1.07 ; non-regulatory Ser96/Ser120/Ser122 flat — confirming the pipeline output in direction, magnitude, and significance. (iii) Total NCC protein flight effect ($+0.089$, 70 peptides) and the matrix-protein effects were confirmed from `protein_effects_gene_anyplex.tsv`. (iv) The RNA recurrence cosine and leave-one-pathway-out range were reproduced from `osd462_rna_recurrence.tsv`. The site-specific pattern (regulatory cluster down, non-regulatory sites flat, total protein flat) is the principal internal control establishing that the phospho-axis result is genuine activating-site regulation rather than a normalization or abundance artifact.

Supplement to Manuscript v8. The spaceflight NCC dephosphorylation phenotype confirmed here was first established by the Cosmic Kidney Disease study (Siew et al., Nature Communications, 2024).